

Basics of System Identification and Realization Theory

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CNRS CRIStAL Lille

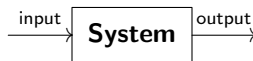
John Leth
Aalborg University

Outline

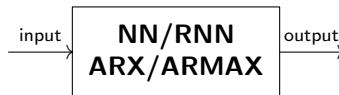
- ▶ Introduction to the identification problem.
- ▶ Linear regression, least-squares estimation.
- ▶ Least-squares estimation of ARX models.
- ▶ State-space estimation, realization theory.

What is system identification ?

We observe the input-output behavior

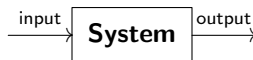


of a (physical) process and try to estimate mathematical models of a certain structure from (time series) data of the system

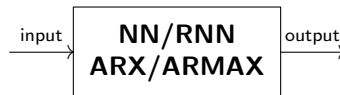


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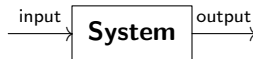
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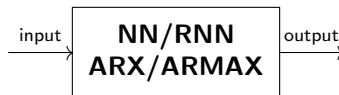
- ▶ System with fundamentally unknown physics
 - ▶ Human behavior
 - ▶ Input: stimuli (visual, auditory, economic signals, etc.)
 - ▶ Output: decisions, reactions

What is system identification ?

We observe the input-output behavior



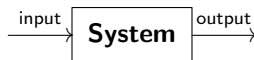
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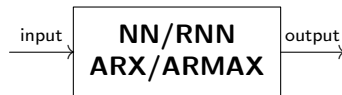
- ▶ Systems with known physics but intractable complexity
 - ▶ Galaxy formation and evolution
 - ▶ Input: initial density fluctuations (early universe), dark matter distribution, etc.
 - ▶ Output: galaxy shapes, sizes, star formation rates

What is system identification ?

We observe the input-output behavior



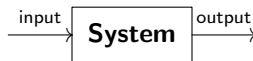
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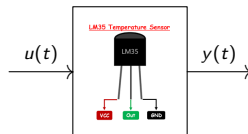
- ▶ Biological systems with many hidden internal states
 - ▶ glucose-insulin dynamics
 - ▶ Input: food intake, insulin injections, etc.
 - ▶ Output: blood glucose levels

What is system identification ?

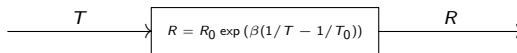
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of a (physical) process



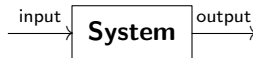
Estimate mathematical models of a certain structure



from (time series) data of the system

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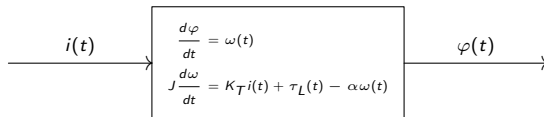
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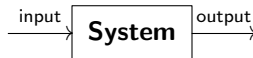
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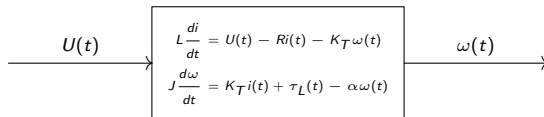
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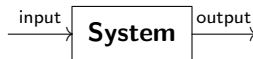
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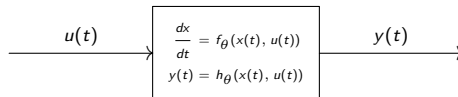
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of a (physical) process



Estimate mathematical models of a certain structure



from (time series) data of the system

Time scale

Measure in discrete-time (with sample time h):

$$t, t + h, t + 2h, t + 3h, \dots \longrightarrow t, t + 1, t + 2, t + 3, \dots$$

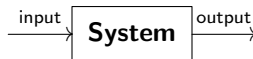
Estimate discrete-time mathematical models e.g., using forward Euler approximation

$$\frac{dx}{dt} = f_{\theta}(x(t), u(t)) \longrightarrow x(t + 1) = x(t) + hf_{\theta}(x(t), u(t))$$

Note equidistant time samples

Simplest case: linear regression

We observe the input-output behavior



Estimate a mathematical model which belongs to the class of **linear regressions**

$$y(t) = \theta^T \phi(t)$$

from data $\{y(t), \phi(t)\}_{t=1}^N$

Linear regression

Model class: $y(t) = \theta^T \phi(t)$

- ▶ $\phi(t) \in \mathbb{R}^m$ regressor (or feature) vector
- ▶ $y(t) \in \mathbb{R}^p$ output vector
- ▶ $\theta^T \in \mathbb{R}^{p \times m}$ unknown parameter to find/estimate

Data: $\{y(t), \phi(t)\}_{t=1}^N$.

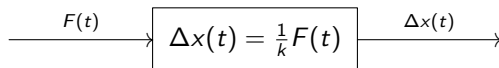
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Data: $\{y(t), \phi(t)\}_{t=1}^N$.

Example (Hooke's law)



- ▶ $\phi(t) = F(t) \in \mathbb{R}$
- ▶ $y(t) = \Delta x(t) \in \mathbb{R}$
- ▶ $\theta^T = \frac{1}{k} \in \mathbb{R}$

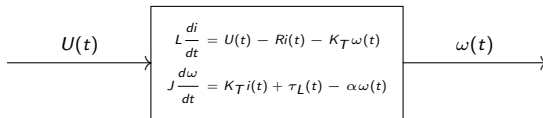
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Data: $\{y(t), \phi(t)\}_{t=1}^N$.

Example (DC motor)



- ▶ $\phi(t) = \left[\frac{di}{dt} \quad i(t) \quad \omega(t) \quad \frac{d\omega}{dt} \quad \tau_L(t) \right]^T \in \mathbb{R}^5$
- ▶ $y(t) = [U(t) \quad 0]^T \in \mathbb{R}^2$
- ▶ $\theta^T = \begin{bmatrix} L & R & K_T & 0 & 0 \\ 0 & K_T & -\alpha & -J & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 5}$

Linear regression

If data $\{y(t), \phi(t)\}_{t=1}^N$ is **assumed** to come from linear regression:

$$y(t) = \theta^T \phi(t), \quad t = 1, \dots, N$$

Reformulate

$$Y_N = \theta^T \Phi_N$$

$$Y_N = [y(1) \quad \cdots \quad y(N)] \quad \text{output matrix}$$

$$\Phi_N = [\phi(1) \quad \cdots \quad \phi(N)] \quad \text{regressor (or data) matrix}$$

Finding $\theta \iff$ solving linear equation $Y_N = \theta^T \Phi_N$

Solving linear equations is easy and well known.

Linear regression

Problem: Data $\{y(t), \phi(t)\}_{t=1}^N$ never satisfy the model (measurement noise, modelling error):

$$y(t) \neq \theta^T \phi(t), \quad t = 1, \dots, N$$

Formally

$$Y_N = \theta^T \Phi_N$$

never has a solution.

We have to find an approximate solution of

$$Y_N = \theta^T \Phi_N$$

Linear regression

Least squares (LS) problem

Find θ such that the (Frobenius) distance between Y_N and $\theta^T \Phi_N$ is minimal

$$\min_{\theta} \|Y_N - \theta^T \Phi_N\|_F^2 = \min_{\theta} \sum_{t=1}^N \|y(t) - \theta^T \phi(t)\|^2$$

Least squares solution

LS solution always exists, there is one solution θ_N if Φ_N is full row rank (the **(finite-sample) persistency of excitation condition**):

$$\theta_N^T = Y_N \Phi_N^T (\Phi_N \Phi_N^T)^{-1}$$

With this θ_N

$$y(t) = \theta_N^T \phi(t) + e_N(t), t = 1, \dots, N$$

and $e_N(t) = y(t) - \theta_N^T \phi(t)$ is the **modelling error**

Linear regression

Least squares (LS) problem

Find θ such that the (Frobenius) distance between Y_N and $\theta^T \Phi_N$ is minimal

$$\min_{\theta} \|Y_N - \theta^T \Phi_N\|_F^2 = \min_{\theta} \sum_{t=1}^N \|y(t) - \theta^T \phi(t)\|^2$$

Note that the LS problem splits into p independent regressions since

$$\sum_{t=1}^N \|y(t) - \theta^T \phi(t)\|^2 = \sum_{i=1}^p \|y_i - \Phi_N^T \theta_i\|^2$$

with $y_i = [y_i(1) \cdots y_i(N)]^T$ the i th column of Y_N^T and θ_i the i th column of θ .

Least squares solution

Hence we are solving p standard LS problems in parallel.

Dealing with modelling error

Core of the problem: the data is not generated by a model of the type $y(t) = \theta^T \phi(t)$.

Solution: consider a model with noise:

$$y(t) = \theta^T \phi(t) + e(t)$$

- ▶ Assume there exists $\theta^T \in \mathbb{R}^{p \times m}$ and a noise sequence $\{e(t)\}_{t=1}^{\infty}$ such that

$$y(t) = \theta^T \phi(t) + e(t), \quad t = 1, 2, \dots$$

- ▶ Assume we can obtain measured data $\{y(t), \phi(t)\}_{t=1}^N$ from this model.
- ▶ Assume that $\{y(t), \phi(t)\}_{t=1}^N$ and $e(t)$ are realizations of stochastic processes

Dealing with modelling error

Question: What is the relationship between the output of the estimated model

$$\hat{y}(t) = \theta_N^T \phi(t), \quad t = 1, 2, \dots$$

and the output of the true system

$$y(t) = \theta^T \phi(t) + e(t), \quad t = 1, 2, \dots$$

Prediction residual: $y(t) - \hat{y}(t)$

Prediction error: $\|y(t) - \hat{y}(t)\|$

We want the model $\hat{y}(t) = \theta_N^T \phi(t)$ to have good predictive power. That is, we want the prediction error to be small, even for data we have not used for estimating θ !

Note that even if $\theta_N = \theta$ (perfect estimation), there is still prediction residual: $y(t) - \hat{y}(t) = e(t) \neq 0$.

Dealing with modelling error

Assume

- ▶ noise is a sample path of an i.i.d stochastic process¹
 $e(t) \sim N(0, \sigma^2 I)$ with $\mathbf{E}(\phi(t)e(t)^T) = 0$ (that is, regressor $\phi(t)$ and noise $e(t)$ are uncorrelated).
- ▶ The empirical covariance $\frac{1}{N}\Phi_N\Phi_N^T$ converges to $\mathbf{E}(\phi(t)\phi(t)^T)$
- ▶ (Asymptotic) Persistency of excitation: $\mathbf{E}(\phi(t)\phi(t)^T)$ is positive definite

With these assumptions:

- ▶ $\lim_{N \rightarrow \infty} \|\theta_N - \theta\|_F^2 = \lim_{N \rightarrow \infty} \sum_{ij} ((\theta_N)_{ij} - \theta_{ij})^2 = 0$
- ▶ $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \|y(t) - \hat{y}(t)\|^2 = \sigma^2$

That is, as $N \rightarrow \infty$, θ_N gets close to θ the average squared prediction error converges to the minimal achievable value, equal to the variance of the noise.

¹By abuse of notation we do not distinguish between a stochastic process and its realizations.

ARX models

The Autoregressive models with exogenous inputs (ARX(p, q)) model is

$$y(t) = - \sum_{i=1}^p a_i y(t-i) + \sum_{j=0}^q b_j u(t-j) + e(t).$$

The one-step-ahead predictor is

$$\hat{y}(t \mid t-1) = - \sum_{i=1}^p a_i y(t-i) + \sum_{j=0}^q b_j u(t-j).$$

$u(t) \in \mathbb{R}$, $y(t) \in \mathbb{R}$ inputs and outputs

$a_1, \dots, a_p \in \mathbb{R}$, $b_0, \dots, b_q \in \mathbb{R}$ – unknown parameter to find/estimate

Estimation of ARX

ARX can be rewritten as

$$y(t) = \theta^T \phi(t) + e(t)$$

and the noise free version (one-step-ahead predictor) can be rewritten as the linear regression

$$y(t) = \theta^T \phi(t) \quad "y(t) = \hat{y}(t | t-1)"$$

with parameter vector

$$\theta = [-a_1 \quad -a_2 \quad \cdots \quad -a_p \quad b_0 \quad b_1 \quad \cdots \quad b_q]^T$$

and regressor (of past outputs, past and current input)

$$\phi(t) = [y(t-1) \quad \cdots \quad y(t-p) \quad u(t) \quad \cdots \quad u(t-q)]^T$$

Estimation of ARX

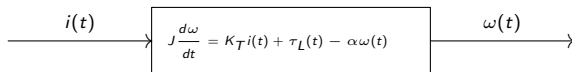
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Example (DC motor)



Estimation of ARX

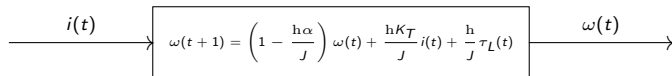
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Example (DC motor)



- ▶ $\phi(t) = [\omega(t-1) \quad i(t-1) \quad \tau_L(t-1)]^T \in \mathbb{R}^3$
- ▶ $y(t) = \omega(t) \in \mathbb{R}$
- ▶ $\theta^T = [1 - \frac{h\alpha}{J} \quad \frac{hK_T}{J} \quad \frac{h}{J}] \in \mathbb{R}^3$

Least squares method for ARX models

Given data $\{y(t), u(t)\}_{t=1}^N$ assumed to come from an $\text{ARX}(p, q)$ system.

Define (as before)

$$Y_N = [y(1) \quad \dots \quad y(N)], \quad \Phi_N = [\phi(1) \quad \dots \quad \phi(N)]$$

The LS problem

$$\min_{\theta} \|Y_N - \theta^T \Phi_N\|_F^2$$

has (as before) a unique solution

$$\theta_N^T = Y_N \Phi_N^T (\Phi_N \Phi_N^T)^{-1}$$

under the persistency of excitation condition (Φ_N full row rank)

Similar analysis as in the linear regression case, but more involved.

Difficulty: $\phi(t)$ depends on $y(t)$.

Least squares method for ARX models

Write

$$\Phi_N = [\phi(1) \quad \dots \quad \phi(N)] = \begin{bmatrix} \mathbf{Y}_{p,N} \\ \mathbf{U}_{q,N} \end{bmatrix}$$

where $\mathbf{Y}_{p,N}$ contains p delayed versions of the output

$$\mathbf{Y}_{p,N} = \begin{bmatrix} y(0) & y(1) & \dots & y(N-1) \\ y(-1) & y(0) & \dots & y(N-2) \\ \vdots & \vdots & & \vdots \\ y(1-p) & y(2-p) & \dots & y(N-p) \end{bmatrix}$$

and the [Hankel \(delay\) matrix](#) $\mathbf{U}_{q,N}$ contains $q+1$ delayed versions of the input

$$\mathbf{U}_{q,N} = \begin{bmatrix} u(1) & u(2) & \dots & u(N) \\ u(0) & u(1) & \dots & u(N-1) \\ \vdots & \vdots & & \vdots \\ u(1-q) & u(2-q) & \dots & u(N-q) \end{bmatrix}$$

Least squares method for ARX models

Write

$$\Phi_N = [\phi(1) \quad \dots \quad \phi(N)] = \begin{bmatrix} \mathbf{Y}_{p,N} \\ \mathbf{U}_{q,N} \end{bmatrix}$$

We can express the (noise free) output

$$y(t) = - \sum_{i=1}^p a_i y(t-i) + \sum_{j=0}^q b_j u(t-j) \quad (1)$$

as a convolution by taking the z-transformed of (1) and assuming that the ARX process is stable²

$$\begin{aligned} y(z) &= \left(1 + \sum_{i=1}^p a_i z^{-i} \right)^{-1} \left(\sum_{j=0}^q b_j z^{-j} \right) u(z) = \left(\sum_{i=0}^{\infty} \alpha_i z^{-i} \right) \left(\sum_{j=0}^q b_j z^{-j} \right) u(z) \\ &= \sum_{k=0}^{\infty} g_k z^{-k} u(z) \end{aligned}$$

with $g_k = \sum_{j=0}^{\min(k,q)} b_j \alpha_{k-j}$ the **impulse response**.

²meaning that the roots of $z^p + \sum_{i=1}^p a_i z^{p-i}$ all lie strictly inside the unit circle.

Least squares method for ARX models

Write

$$\Phi_N = [\phi(1) \quad \dots \quad \phi(N)] = \begin{bmatrix} \mathbf{Y}_{p,N} \\ \mathbf{U}_{q,N} \end{bmatrix}$$

We can express the (noise free) output as a convolution

$$y(t) = - \sum_{i=1}^p a_i y(t-i) + \sum_{j=0}^q b_j u(t-j) = \sum_{k=0}^{\infty} g_k u(t-k) \quad (1)$$

where $\{g_k\}$ is the [impulse response](#).

Thus, each row of $\mathbf{Y}_{p,N}$ is a [linear combination](#) of delayed input sequences and therefore

$$\mathbf{Y}_{p,N} = \mathbf{G} \mathbf{U}_{\infty,N}.$$

with matrix $\mathbf{G} = [G_{ij}]$ having a Toeplitz-like structure determined by the impulse response $G_{ij} = g_{j-i-1}$, $i = 1, \dots, p$, $j > i$ for the non-zero elements

Least squares method for ARX models

Combining everything:

$$\Phi_N = [\phi(1) \quad \dots \quad \phi(N)] = \begin{bmatrix} \mathbf{Y}_{p,N} \\ \mathbf{U}_{q,N} \end{bmatrix} = T \mathbf{U}_{\infty,N}, \quad T = \begin{bmatrix} G \\ S \end{bmatrix}.$$

where matrix $S = [I \ 0]$ extracts the first $q + 1$ delays.

Rank Implications: We obtain

$$\text{rank}(\Phi_N) \leq \min(\text{rank}(T), \text{rank}(\mathbf{U}_{\infty,N})).$$

To ensure full row rank $p + q + 1$, the following must hold:²

1. $\text{rank}(\mathbf{U}_{\infty,N}) = p + q + 1$ (persistent excitation of inputs)
2. $\dim(\text{Im}(\mathbf{U}_{\infty,N}) \cap \ker(T)) = 0$

²in general: $p + q + 1 \leq \text{rank}(\mathbf{U}_{\infty,N}) < \infty$ and
 $\dim(\text{Im}(\mathbf{U}_{\infty,N}) \cap \ker(T)) + p + q + 1 = \text{rank}(\mathbf{U}_{\infty,N})$

Exercise

- ▶ Generate data $\{y(t), u(t)\}_{t=1}^N$ from

$$y(t) = -a_1 y(t-1) + b_0 u(t) + b_1 u(t-1) + e(t)$$

with $e(t)$, $u(t)$ white noise (variance one), $-a_1 = 0.3$, $b_0 = 1$ and $b_1 = -2$

- ▶ Use the data to estimate the coefficients a_1 , b_0 and b_1 using least squares.
- ▶ Change the variance of $u(t)$ to 5, what happens?
- ▶ Change $-a_1$ to 3, what happens?

Estimation ARX models: identifiability

If $\{y(t), u(t)\}_{t=1}^{\infty}$ satisfies

$$y(t) = 3y(t-1) + u(t) - 2u(t-1) \quad (1)$$

it also satisfies the following two equations

$$y(t) = 4y(t-1) - 3y(t-2) + u(t) - 3u(t-1) + 2u(t-2) \quad (2)$$

$$y(t) = 8y(t-1) - 15y(t-2) + u(t) - 7u(t-1) + 10u(t-2) \quad (3)$$

How to distinguish (2) and (3) ?

Two parameters $\theta_1 = (4, -3, 1, -3, 2)$ and $\theta_2 = (8, -15, 1, -7, 10)$ fit the data, as $y(t) = \theta_i^T \phi(t)$, $i = 1, 2$.

Estimation ARX models: identifiability

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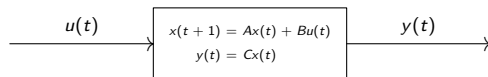
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Exercise 1

- ▶ Verify that (1) implies (2) and (3)
- ▶ With initial data $y(-1) = u(-1) = 0$ and (impulse) input $u(0) = 1$ and $u(t) = 0$, $t \geq 1$, investigate if (2) and (3) can be distinguished by computing the impulse responses.

LTI state-space models

Linear Time Invariant (LTI) state-space models/systems

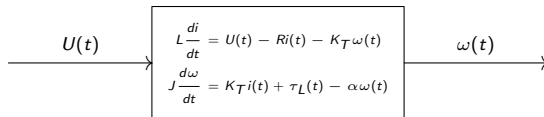


$u(t) \in \mathbb{R}^m$, $y(t) \in \mathbb{R}^p$ inputs and outputs

$x(t) \in \mathbb{R}^n$ state, $x(0) = 0$.

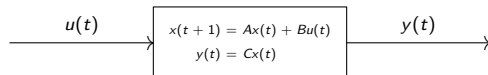
$A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$ – unknown parameters to find/estimate

Example (DC motor)



LTI state-space models

Linear Time Invariant (LTI) state-space models/systems

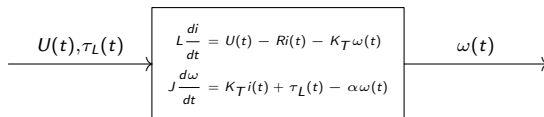


$u(t) \in \mathbb{R}^m$, $y(t) \in \mathbb{R}^p$ inputs and outputs

$x(t) \in \mathbb{R}^n$ state, $x(0) = 0$.

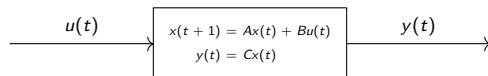
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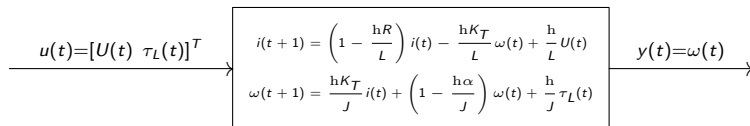


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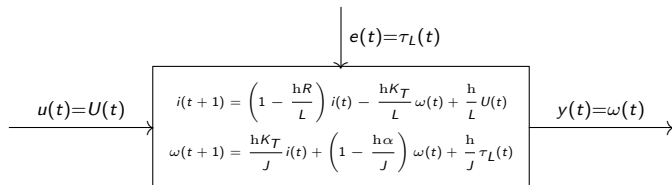
Example (DC motor)



$$\blacktriangleright x(t) = \begin{bmatrix} i(t) \\ \omega(t) \end{bmatrix}, A = \begin{bmatrix} 1 - \frac{hR}{L} & -\frac{hK_T}{L} \\ \frac{hK_T}{J} & 1 - \frac{h\alpha}{J} \end{bmatrix}, B = \begin{bmatrix} \frac{h}{L} & 0 \\ 0 & \frac{h}{J} \end{bmatrix}, C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

LTI state-space models

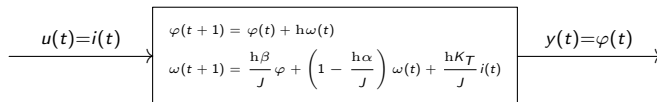
Example (DC motor)



- ▶ $x(t+1) = Ax(t) + Bu(t) + Ee(t)$, $y(t) = Cx(t)$
- ▶ $e(t) \in \mathbb{R}^l$ exogenous input
- ▶ $E \in \mathbb{R}^{n \times l}$ additional unknown parameter to find/estimate
- ▶ $x(t) = \begin{bmatrix} i(t) \\ \omega(t) \end{bmatrix}$, $A = \begin{bmatrix} 1 - \frac{hR}{L} & -\frac{hK_T}{L} \\ \frac{hK_T}{J} & 1 - \frac{h\alpha}{J} \end{bmatrix}$, $B = \begin{bmatrix} \frac{h}{L} \\ 0 \end{bmatrix}$, $E = \begin{bmatrix} 0 \\ \frac{h}{J} \end{bmatrix}$,
 $C = \begin{bmatrix} 0 & 1 \end{bmatrix}$

LTI state-space models

Example (DC motor)



- ▶ $x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t)$
- ▶ $\tau_L(t) = \beta \sin(\varphi) \approx \beta\varphi$ exogenous input
- ▶ $x(t) = \begin{bmatrix} \varphi(t) \\ \omega(t) \end{bmatrix}, \quad A = \begin{bmatrix} 1 & h \\ \frac{h\beta}{J} & 1 - \frac{h\alpha}{J} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{hK_T}{J} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output), and consider the state space system

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t) \quad (4)$$

Using the z-transformed the input output behaviour of the state space system can be described by the transfer function

$$y(z) = G(z)u(z)$$

with

$$\begin{aligned} G(z) &= C(zI - A)^{-1}B \\ &= C \frac{\text{adj}(zI - A)}{\det(zI - A)} B \\ &= \frac{\alpha_{n-1}z^{n-1} + \alpha_{n-2}z^{n-2} + \dots + \alpha_1z + \alpha_0}{z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0} \end{aligned}$$

The coefficients of G can be determined as follows

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output), and consider the state space system

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t) \quad (4)$$

Let

$$\chi_A(z) = \det(zI - A) = z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0$$

denote the characteristic polynomial of A

Define the **reachability** matrix

$$[A|B] = [B \quad AB \quad \dots \quad A^{n-1}B]$$

and if $\text{rank}([A|B]) = n$ the coordinate transform

$$T_r = [A|B]T_a^{-1}, \quad T_a = \begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_{n-1} & 1 \\ a_2 & a_3 & \dots & a_{n-1} & 1 & 0 \\ a_3 & \dots & a_{n-1} & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n-1} & 1 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 & 0 \end{bmatrix}$$

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output), and consider the state space system

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t) \quad (4)$$

Using $\bar{x} = T_r x$ the state space system (4) is transformed into **control canonical form**

$$\bar{x}(t+1) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \bar{x}(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [\bar{c}_0 \quad \bar{c}_1 \quad \bar{c}_2 \quad \cdots \quad \bar{c}_{n-1}] \bar{x}(t)$$

And therefore

$$G(z) = C \frac{\text{adj}(zI - A)}{\det(zI - A)} B = \frac{\bar{c}_{n-1}z^{n-1} + \bar{c}_{n-2}z^{n-2} + \cdots + \bar{c}_1z + \bar{c}_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0}$$

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Define the **observability** matrix

$$(A|C) = [A^T | C^T]^T$$

and if $\text{rank}(A|C) = \text{rank}([A^T | C^T]) = n$ the coordinate transform

$$T_o = ([A^T | C^T] T_a^{-1})^{-T}, \quad T_a = \begin{bmatrix} a_1 & a_2 & a_3 & \cdots & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & a_{n-1} & 1 & 0 \\ a_3 & \cdots & a_{n-1} & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n-1} & 1 & 0 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output), and consider the state space system

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t) \quad (4)$$

Using $\bar{x} = T_o x$ the state space system (4) is transformed into **observable canonical form**

$$\bar{x}(t+1) = \begin{bmatrix} 0 & 0 & 0 & \cdots & -a_0 \\ 1 & 0 & 0 & \cdots & -a_1 \\ 0 & 1 & \vdots & \vdots & -a_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix} \bar{x}(t) + \begin{bmatrix} \bar{b}_0 \\ \bar{b}_1 \\ \bar{b}_2 \\ \vdots \\ \bar{b}_{n-1} \end{bmatrix} u(t)$$

$$y(t) = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \bar{x}(t)$$

And therefore

$$G(z) = C \frac{\text{adj}(zI - A)}{\det(zI - A)} B = \frac{\bar{b}_{n-1}z^{n-1} + \bar{b}_{n-2}z^{n-2} + \cdots + \bar{b}_1z + \bar{b}_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0}$$

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output)

$\{y(t), u(t), x(t)\}_{t=1}^{\infty}$ satisfies a state-space model

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t)$$

$\iff \{u(z), y(z)\}_{t=1}^{\infty}$ is the input-output pair of a transfer function

$$y(z) = G(z)u(z)$$

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and $\{u(z), y(z)\}_{t=1}^{\infty}$ the z-transformed of the data

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and $\{u(z), y(z)\}_{t=1}^{\infty}$ the z-transformed of the data

$\iff \{y(t), u(t)\}_{t=1}^{\infty}$ satisfy a (noise free) ARX model

$$y(t) = - \sum_{i=1}^n a_{n-i}y(t-i) + \sum_{i=1}^n \bar{b}_{n-i}u(t-i)$$

ARX and LTI state-space models

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The term $b_0 u(t)$ occurs in the ARX model if the output equation contains a direct term: $y(t) = Cx(t) + b_0 u(t)$

$p, m > 1$ MIMO (multi input multi output) $\implies$ relationship between state-space and ARX is complicated.

ARX and LTI state-space models

Assume $p = m = 1$ SISO (single input single output)

$\{y(t), u(t), x(t)\}_{t=1}^{\infty}$ satisfies a state-space model

$$x(t+1) = Ax(t) + Bu(t), \quad y(t) = Cx(t)$$

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$$y(t) = - \sum_{i=1}^n a_{n-i} y(t-i) + \sum_{i=1}^n \bar{b}_{n-i} u(t-i)$$

Exercise 2

- ▶ Compute the transfer functions corresponding to the two ARX(2,2) models in exercise 1 and use these to conclude that they have the same input-output behaviour.
- ▶ Compute the state space representations corresponding to the two ARX(2,2) models in exercise 1 (note the direct term).

Motivating example

Parametrization for $\theta = [\theta_1 \ \theta_2 \ \theta_3]^T$:

$$\begin{aligned}x(t+1) &= \begin{bmatrix} 0 & 1 \\ \theta_1 & \theta_2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= [\theta_3 \quad 1] x(t)\end{aligned}$$

$\theta = [2 \ -1 \ 2]^T$ and $\theta' = [-2 \ 3 \ -2]^T$ give the same data $\{y(t), u(t)\}_{t=1}^{\infty}$.

Motivating example

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$\theta = [2 \ -1 \ 2]^T$ and $\theta' = [-2 \ 3 \ -2]^T$ give the same data $\{y(t), u(t)\}_{t=1}^{\infty}$.

The non-uniqueness is caused by pole-zero cancellation

$$\begin{aligned}G_{\theta}(z) &= \frac{z+2}{z^2+z-2} = \frac{1}{z-1} \\ G_{\theta'}(z) &= \frac{z-2}{z^2-3z+2} = \frac{1}{z-1}\end{aligned}$$

Motivating example

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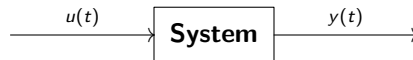
Non-uniqueness creates

- ▶ pole-zero cancellation - potential problem since the pole and zero might not be identical in the true system
- ▶ potential state explosion of hidden modes

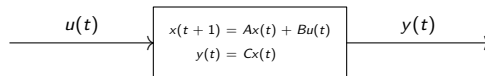
For identification of state-space models we need realization theory.

Realization theory

For the input-output system



which state-space model



can describe the observed behavior of the system ?

Input-output map of a LTI state-space representation

$$\Sigma : \begin{cases} x(t+1) = Ax(t) + Bu(t), & x(0) = 0 \\ y(t) = Cx(t) \end{cases}$$

$$A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times m}, C \in \mathbb{R}^{p \times n}.$$

dimension of Σ = number of state variables n .

The corresponding input-output map Y_Σ maps input $u = u(\cdot)$ to output $y = y(\cdot)$, using initial state $x(0) = 0$: $y = Y_\Sigma(u)$ with

$$Y_\Sigma(u)(t) = y(t) = \sum_{s=0}^{t-1} CA^{(t-s-1)}Bu(s)$$

Σ is a **realization** of a specified input-output map $Y : u(\cdot) \mapsto y(\cdot)$, iff $Y_\Sigma = Y$.

Realization problem

For a specified input-output map Y find a (preferably minimal) linear system Σ such that Σ realizes Y .

Impulse response

A potential input-output map of a linear system is determined by its impulse response:

Impulse response $G(t)$

$$y(t) = Y(u)(t) = \sum_{s=0}^{t-1} G(t-s)u(s)$$

Σ is a realization, iff

$$G(t) = CA^{t-1}B$$

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We note that using initial condition $x(0) = 0$ implies that we only sum over past inputs $u(0), \dots, u(t-1)$ this should be interpreted with the usual convention that the input is zero for negative times:

$$y(t) = \sum_{s=0}^{t-1} G(t-s)u(s) = \sum_{k=1}^t G(k)u(t-k) = \sum_{k=1}^{\infty} G(k)u(t-k)$$

Impulse response

A potential input-output map of a linear system is determined by its impulse response:

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Exercise 3

- Use the computed state space representations from exercise 2 to calculate the corresponding impulse responses.

Markov parameters

Markov parameters

$$M_k = G(k+1)$$

Classical step. Σ is a realization of $Y \iff M_k = CA^k B$

Observability and reachability

State-space representation

$$\Sigma : \begin{cases} x(t+1) = Ax(t) + Bu(t), & x(0) = 0 \\ y(t) = Cx(t) \end{cases}$$

is **reachable**, if for any $z \in \mathbb{R}^n$, there exists $u(\cdot)$ and T s.t. $z = x(T)$.

observable, if for any $z_1 \neq z_2 \in \mathbb{R}^n$, there exists $u(\cdot)$ and T s.t.

$$x(t+1) = Ax(t) + Bu(t), \quad x(0) = z_1, \quad y(t) = Cx(t)$$

$$x'(t+1) = Ax'(t) + Bu(t), \quad x'(0) = z_2, \quad y'(t) = Cx'(t)$$

$$y(T) \neq y'(T)$$

Observability and reachability

For the state-space representation

$$\Sigma : \begin{cases} x(t+1) = Ax(t) + Bu(t), & x(0) = 0 \\ y(t) = Cx(t) \end{cases}$$

The following 3 items are equivalent

- ▶ Σ reachable
- ▶ $\text{Rank}[A|B] = n$
- ▶ $\text{Span}\{A^k Bu \mid k \geq 0, u \in \mathbb{R}^m\} = \mathbb{R}^n$

The following 4 items are equivalent

- ▶ Σ observable
- ▶ $\ker(A|C) = \{0\}$
- ▶ $\text{Rank}(A|C) = \text{Rank}[A^T|C^T] = n$
- ▶ $\bigcap_{k=0}^{\infty} \ker CA^k = \{0\}$

Minimality

A minimal realization of an input–output system is a state-space representation that reproduces the same input–output behavior using the smallest possible number of state variables.

Theorem 1 (Minimality)

1. Σ is a minimal realization of $Y \iff \Sigma$ is *reachable*, and Σ is *observable*.
2. If (A, B, C) and $(\hat{A}, \hat{B}, \hat{C})$ are minimal realizations of Y , then they are isomorphic: there exists a nonsingular matrix T s.t.:

$$TAT^{-1} = \hat{A}, \quad TB = \hat{B}, \quad CT^{-1} = \hat{C}.$$

3. Any realization Σ of Y , can be converted to a minimal realization with the same input-output map Y .

Minimality

Theorem 1 (Minimality)

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3. Any realization Σ of Y , can be converted to a minimal realization with the same input-output map Y .

Exercise 4

- Verify that the computed state space representations from exercise 2 are non minimal, and find a minimal realization.

Reachability reduction

Basis $b_1, \dots, b_n$ such that $b_1, \dots, b_r$ is a basis for

$$\text{Im}[A|B]$$

In the new basis³

$$A = \begin{bmatrix} A_r & \star \\ 0 & A_{ur} \end{bmatrix}, \quad B = \begin{bmatrix} B_r \\ 0 \end{bmatrix}, \quad C = [C_r \quad \star]$$

(A_r, B_r, C_r) is reachable and realizes the same input-output map as (A, B, C) .

³using the coordinate transformation $z = T_r x$ with $T_r = [b_1 \cdots b_n]^{-1}$

Observability reduction

Basis $b_1, \dots, b_n$ such that $b_{o+1}, \dots, b_n$ is a basis for

$$\ker(A|C)$$

In the new basis⁴

$$A = \begin{bmatrix} A_o & 0 \\ \star & A_{uo} \end{bmatrix}, \quad B = \begin{bmatrix} B_o \\ \star \end{bmatrix}, \quad C = [C_o, \quad 0]$$

(A_o, B_o, C_o) is observable and realizes the same input-output map as (A, B, C) .

⁴using the coordinate transformation $z = T_o x$ with $T_o = [b_1 \cdots b_n]^{-1}$

Transforming an LTI to a minimal one

Minimization procedure

1. Transform (A, B, C) to a reachable (A_r, B_r, C_r) with the same input-output map.
 2. Transform (A_r, B_r, C_r) to an observable (A_m, B_m, C_m) with the same input-output map.
- ▶ (A_m, B_m, C_m) is reachable and observable, realizes the same input-output map as (A, B, C) .
 - ▶ Dimension of $(A_m, B_m, C_m) \leq$ dimension of (A, B, C) .

Transforming an LTI to a minimal one

Minimization procedure

1. Transform (A, B, C) to a reachable (A_r, B_r, C_r) with the same input-output map.
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- ▶ (A_m, B_m, C_m) is reachable and observable, realizes the same input-output map as (A, B, C) .
 - ▶ Dimension of $(A_m, B_m, C_m) \leq$ dimension of (A, B, C) .

Exercise 5

- ▶ Use the above procedure on one of the computed state space representations from exercise 2.

Isomorphism of reachable and observable LTIs

(A, B, C) and $(\tilde{A}, \tilde{B}, \tilde{C})$ are reachable and observable realizations of Y .

n dimension of (A, B, C) , $\tilde{n}$ dimension of $(\tilde{A}, \tilde{B}, \tilde{C})$.

(A, B, C) and $(\tilde{A}, \tilde{B}, \tilde{C})$ realizations of $Y \implies$

$$CA^k B = M_k = \tilde{C} \tilde{A}^k \tilde{B}$$

Consider k -step observability/reachability matrices:

$$O_k = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^k \end{bmatrix} \quad \tilde{O}_k = \begin{bmatrix} \tilde{C} \\ \tilde{C} \tilde{A} \\ \vdots \\ \tilde{C} \tilde{A}^k \end{bmatrix}$$

$$R_k = \begin{bmatrix} B & AB & \cdots & A^k B \end{bmatrix}, \quad \tilde{R}_k = \begin{bmatrix} \tilde{B} & \tilde{A} \tilde{B} & \cdots & \tilde{A}^k \tilde{B} \end{bmatrix},$$

Isomorphism of reachable and observable LTIs

Take SISO (the MIMO case is similar).

Markov-parameters equal $\implies$

$$O_k R_l = \begin{bmatrix} M_0 & \dots & M_l \\ M_1 & \dots & M_{l+1} \\ \vdots & \dots & \vdots \\ M_k & \dots & M_{k+l} \end{bmatrix} = \tilde{O}_k \tilde{R}_l$$

Assume $n \geq \tilde{n}$:

$$n = \text{Rank } O_{n-1} R_{n-1} = \text{Rank } \tilde{O}_{n-1} \tilde{R}_{n-1} = \tilde{n}$$

Assuming $n \leq \tilde{n}$ give the same result i.e., $n = \tilde{n}$.

Isomorphism of reachable and observable LTIs

Define:

$$T = \tilde{O}_{n-1}^{-1} O_{n-1}$$

T is an isomorphism from (A, B, C) to $(\tilde{A}, \tilde{B}, \tilde{C})$:

$$TAT^{-1} = \tilde{A}, \quad TB = \tilde{B}, \quad CT^{-1} = \tilde{C}$$

Isomorphism of reachable and observable LTIs

For all $l \geq 0$:

$$O_{n-1}R_l = \tilde{O}_{n-1}\tilde{R}_l \implies TR_l = \tilde{R}_l,$$

Since $R_0 = B$, $\tilde{R}_0 = \tilde{B}$:

$$TB = \tilde{B}$$

Moreover, with $R_l^+ = AR_{l-1}$ and $\tilde{R}_l^+ = \tilde{A}\tilde{R}_{l-1}$ the above also holds true, and

$$TAR_{n-1} = TR_n^+ = \tilde{R}_n^+ = \tilde{A}\tilde{R}_{n-1} = \tilde{A}TR_{n-1} \implies$$

$$TA = \tilde{A}T \iff TAT^{-1} = \tilde{A}$$

Isomorphism of reachable and observable LTIs

Since

$$\tilde{O}_{n-1}T = \tilde{O}_{n-1}\tilde{O}_{n-1}^{-1}O_{n-1} = O_{n-1}$$

that is

$$\begin{bmatrix} \tilde{C}T \\ \tilde{C}\tilde{A}T \\ \vdots \\ \tilde{C}\tilde{A}^{n-1}T \end{bmatrix} = \tilde{O}_{n-1}T = O_{n-1} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

and therefore

$$\tilde{C}T = C \implies CT^{-1} = \tilde{C}$$

Existence of a realization

Recall $M_k = G(k+1)$ – Markov parameters

Hankel matrix of Y

$$H_Y = \begin{bmatrix} M_0 & M_1 & M_2 & \cdots \\ M_1 & M_2 & M_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Theorem 2

- ▶ Y has a realization by an LTI $\iff \text{Rank} H_Y < +\infty$.
- ▶ $\text{Rank} H_Y$ is the dimension of a minimal realization of Y .

Hence Y has no finite dimensional realization if $\text{Rank} H_Y < +\infty$ e.g., $G(t) = 1/t$.

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Exercise 6

- ▶ Apply the theorem to $Y = Y_\Sigma$ with Σ the computed state space representations from exercise 2.

Existence of a realization: idea of the proof

For simplicity: assume SISO

$\mathbb{R}^\infty$ – the set of all infinite vectors $\begin{bmatrix} z_0 \\ z_1 \\ \vdots \end{bmatrix}$.

$\text{Im}H_Y \subseteq \mathbb{R}^\infty$ – the linear space generated by the columns of H_Y .

$\text{Rank}H_Y = \dim \text{Im}H_Y$.

Idea of the proof:

- ▶ (A, B, C) is a realization of Y of dimension $n \implies \dim \text{Im}H_Y \leq n$.
- ▶ $\dim \text{Im}H_Y < +\infty \implies$ we can construct a realization (A, B, C) of Y .

Existence of a realization: proof of necessity

(A, B, C) is a realization of $Y \implies$

$$\begin{bmatrix} M_k \\ M_{k+1} \\ \vdots \\ M_{k+i} \\ \vdots \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^i \\ \vdots \end{bmatrix} A^k B$$

Define the linear map

$$\mathcal{O} : \mathbb{R}^n \rightarrow \mathbb{R}^\infty; x \mapsto \begin{bmatrix} C \\ CA \\ \vdots \\ CA^i \\ \vdots \end{bmatrix} x$$

$$(\dim \operatorname{Im} \mathcal{O} \leq n \text{ and } \operatorname{Im} H_Y \subseteq \operatorname{Im} \mathcal{O}) \implies \dim \operatorname{Im} H_Y \leq n.$$

Existence of a realization: proof of sufficiency

Assume $\mathcal{X} := \text{Im} H_Y$ finite dimensional.

$\mathcal{B}$ – first column of H_Y

Projection operator:

$$\mathcal{C} : \mathbb{R}^\infty \rightarrow \mathbb{R}; \begin{bmatrix} z_0 \\ z_1 \\ \vdots \end{bmatrix} \mapsto z_0$$

Shift operator:

$$\mathcal{A} : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty; \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_i \\ \vdots \end{bmatrix} \mapsto \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_{i+1} \\ \vdots \end{bmatrix}$$

$\mathcal{A}, \mathcal{C}$ linear maps

Existence of a realization: proof of sufficiency

Notice: all columns of H_Y are shifted versions of the first column

$$\begin{bmatrix} M_k \\ M_{k+1} \\ \vdots \\ M_{k+i} \\ \vdots \end{bmatrix} = \mathcal{A}^k \begin{bmatrix} M_0 \\ M_1 \\ \vdots \\ M_i \\ \vdots \end{bmatrix} = \mathcal{A}^k \mathcal{B}$$

$$\mathcal{A}\mathcal{X} \subseteq \mathcal{X}$$

Existence of a realization: proof of sufficiency

Choose a basis of $\mathcal{X}$, and let

- ▶ B the coordinates of $\mathcal{B} \in \mathcal{X}$ in this basis
- ▶ A the matrix representation of the linear map

$$\mathcal{X} \ni x \mapsto \mathcal{A}x \in \mathcal{X}$$

- ▶ C the matrix representation of the linear map

$$\mathcal{X} \ni x \mapsto \mathcal{C}x \in \mathbb{R}$$

$$M_k = \mathcal{C}\mathcal{A}^k\mathcal{B} = CA^kB,$$

i.e. (A, B, C) is a realization of Y .

Ho-Kalman algorithm

1. Find a factorization

$$H_{N,N+1} = \begin{bmatrix} M_0 & M_1 & \cdots & M_N \\ M_1 & M_2 & \cdots & M_{N+1} \\ \vdots & \vdots & \vdots & \vdots \\ M_{N-1} & M_N & \cdots & M_{2N-1} \end{bmatrix} = OR$$

s.t. O full column rank, R full row rank.

(e.g, SVD: $H_{N,N+1} = U\Sigma V^T$, $O = U\Sigma^{1/2}$, $R = \Sigma^{1/2}V^T$).

$$2. R = [R_1 \quad R_2 \quad \cdots \quad R_{N+1}], O = \begin{bmatrix} O_1 \\ O_2 \\ \vdots \\ O_N \end{bmatrix}.$$

3. $B = R_1$, $C = O_1$, and A solves

$$A [R_1 \quad R_2 \quad \cdots \quad R_N] = [R_2 \quad R_3 \quad \cdots \quad R_{N+1}]$$

Ho-Kalman algorithm

Exercise 7

For each $i = 1, 2$, consider

$x(t+1) = A_i x(t) + Bu(t)$, $y(t) = Cx(t)$, $x(0) = 0$, where

$$B = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, C = \begin{bmatrix} \frac{3}{2} & 6 & -\frac{7}{2} & \frac{3}{2} \end{bmatrix},$$

$$A_1 = \begin{bmatrix} \frac{1}{10} & -\frac{2}{5} & \frac{2}{5} & 0 \\ \frac{9}{40} & -\frac{1}{5} & \frac{9}{40} & -\frac{9}{40} \\ \frac{40}{9} & -\frac{5}{7} & \frac{40}{29} & -\frac{40}{9} \\ \frac{40}{3} & -\frac{10}{11} & \frac{40}{7} & \frac{1}{40} \end{bmatrix}, \quad A_2 = \begin{bmatrix} \frac{6}{5} & \frac{7}{10} & -\frac{7}{10} & 0 \\ -\frac{79}{8} & 20 & -\frac{79}{8} & \frac{79}{8} \\ -\frac{79}{8} & \frac{39}{2} & -\frac{75}{8} & \frac{79}{8} \\ -\frac{357}{40} & \frac{101}{5} & -\frac{413}{40} & \frac{81}{8} \end{bmatrix}.$$

Compute $M_k = CA_i^k B$. For $N = 2, 3, 4, 5$, form $H_{N,N+1}$ and apply the Ho-Kalman algorithm⁵ to obtain $(\hat{A}, \hat{B}, \hat{C})$. Verify that $\hat{M}_k = \hat{C}\hat{A}^k\hat{B}$ matches M_k .

⁵Use a tolerance of 10^{-10} for the singular values.

Correctness of Ho-Kalman algorithm and partial realization

$$H_{N,N} = \begin{bmatrix} M_0 & M_1 & \cdots & M_{N-1} \\ M_1 & M_2 & \cdots & M_N \\ \vdots & \vdots & \vdots & \vdots \\ M_{N-1} & M_N & \cdots & M_{2N-2} \end{bmatrix}$$

$$H_{N+1,N} = \begin{bmatrix} M_0 & M_1 & \cdots & M_{N-1} \\ M_1 & M_2 & \cdots & M_N \\ \vdots & \vdots & \vdots & \vdots \\ M_N & M_{N+1} & \cdots & M_{2N-1} \end{bmatrix}$$

Correctness of Ho-Kalman algorithm and partial realization

Theorem 3 (Ho-Kalman algorithm & partial realization)

1. $\text{Rank} H_{N,N} = \text{Rank} H_Y \implies (A, B, C)$ is a minimal realization of Y
2. If Y has a realization of dimension less than N , then $\text{Rank} H_{N,N} = \text{Rank} H_Y$.
3. $\text{Rank} H_{N,N} = \text{Rank} H_{N+1,N} = \text{Rank} H_{N,N+1} \implies (A, B, C)$ is a so called $2N$ realization of Y , i.e.

$$M_k = CA^k B, \quad k = 0, 1, \dots, 2N - 1$$

Intuition behind the theorem

(A, B, C) is a realization of $Y \implies$

$$H_{N,N} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{N-1} \end{bmatrix}}_O \underbrace{\begin{bmatrix} B & AB & \dots & A^{N-1}B \end{bmatrix}}_R$$

$N \geq \text{state-space dimension} \implies O$ and R are full rank.

Identification of LTI state-space models via realization algorithm

We do not have access to the Markov parameters, we only have access to data.

- ▶ Estimate finitely many Markov parameters $\{M_k\}_{k=0}^{2N}$ from

$$y(t) = \sum_{s=0}^{t-1} G(t-s)u(s) = \sum_{s=0}^{t-1} M_{t-1-s}u(s) = \sum_{k=0}^{t-1} M_k u(t-1-k)$$

- ▶ Construct the Hankel matrix $H_{N,N+1}$ from the estimated Markov parameters
- ▶ Apply HO-Kalman realization algorithm.

Better algorithms exist, this one is only for illustration !

Estimating Markov parameters: linear regression

First option: use linear regression, by assuming that for $k \geq K$, $M_k \approx 0$.

For stable systems $M_k \approx 0$ for large enough k .

$$y(t+K) = \underbrace{\begin{bmatrix} M_0 & \dots & M_{K-1} \end{bmatrix}}_{\theta^T} \underbrace{\begin{bmatrix} u(t+K-1) \\ u(t+K-2) \\ \vdots \\ u(t) \end{bmatrix}}_{\phi(t)}$$

$$y(t+K) = \theta^T \phi(t)$$

classical linear regression problem !

Estimating Markov parameters: linear regression

$$y(t + K) = \theta^T \phi(t)$$

has unique solution,

$$\begin{bmatrix} M_0 & \cdots & M_{K-1} \end{bmatrix} = \theta^T = Y_N \Phi_N^T (\Phi_N \Phi_N^T)^{-1}$$

$$Y_N = \begin{bmatrix} y(K) & \cdots & y(N + K) \end{bmatrix}$$

$$\Phi_N = \begin{bmatrix} \phi(0) & \cdots & \phi(N) \end{bmatrix} = \begin{bmatrix} u(K-1) & u(K) & \cdots & u(N+K-1) \\ u(K-2) & u(K-1) & \cdots & u(N+K-2) \\ \vdots & \vdots & \cdots & \vdots \\ u(0) & u(1) & \cdots & u(N-1) \end{bmatrix}$$

if Φ_N is full row rank, i.e., u is **persistently exciting of order K**

persistently exciting of order $K \implies$ we can recover K Markov parameters !

Estimating Markov parameters: linear regression

Exercise 8

Let $K = 4$ and simulate N data points from the stable state-space system in exercise 7 with the following inputs

1. $u(t) \equiv 1, \quad t = 0, 1, 2, \dots$
2. $u(t) = \sin\left(\frac{2\pi t}{4}\right), \quad t = 0, 1, 2, \dots$
3. $u(t) = \sin\left(\frac{2\pi t}{5}\right) + 0.7 \sin\left(\frac{2\pi t}{7}\right), \quad t = 0, 1, 2, \dots$

Estimating Markov parameters using linear regression (you will run into problems when using item 1 and 2, why?)

Estimating Markov parameters: statistics

If u is white noise:

$$M_k = E[y(t)u(t-k-1)] \approx \frac{1}{N} \sum_{t=k}^N y(t)u(t-k-1)$$

- ▶ for an LTI system, output is a weighted sum of past inputs,
- ▶ if the input is white noise, different lags of the input are uncorrelated,
- ▶ therefore the cross-correlation between output and lagged input isolates exactly one impulse-response coefficient / Markov parameter.

Estimating Markov parameters: statistics

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Exercise 9

Repeat exercise 8 with u being white noise

Estimating Markov parameters: statistics

General u :

$$\underbrace{E[y(K)u(K-k-1)]}_{\Lambda_k} = \underbrace{[M_0 \quad \cdots \quad M_{K-1}]}_{\theta^T} \underbrace{\begin{bmatrix} R_k \\ R_{k-1} \\ \vdots \\ R_0 \\ \vdots \\ R_{K-k-1} \end{bmatrix}}_{\psi_k}$$

$$R_k = E[u(t)u(t-k)] \approx \frac{1}{N} \sum_{t=k}^N u(t)u(t-k)$$

- ▶ Λ_k is an output-input cross-correlation,
- ▶ $R_k = R_{-k}$ is an input autocorrelation,
- ▶ together they form linear equations whose unknowns are the Markov parameters

Estimating Markov parameters: statistics

Derivation

$$\begin{aligned} E[y(t)u(t-\ell)] &= E \left[\sum_{s=0}^{t-1} M_s u(t-s-1)u(t-\ell) \right] \\ &= \sum_{s=0}^{t-1} M_s E[u(t-s-1)u(t-\ell)] \\ &= \sum_{s=0}^{t-1} M_s R_{\ell-s-1} \end{aligned}$$

In particular, the case of white noise input for $\ell = k + 1$ yields $R_{k+1-s-1} = 1$ for $s = k$ and zero otherwise (we assume unit variance for simplicity)

Estimating Markov parameters: statistics

Notice:

$$\Lambda_k = \theta^T \psi_k$$

linear regression !

Linear regression has a unique solution, if

$$\begin{bmatrix} \psi_0 & \dots & \psi_{K-1} \end{bmatrix} = \begin{bmatrix} R_0 & R_1 & \dots & R_{K-1} \\ R_1 & R_0 & \dots & R_{K-2} \\ \vdots & \vdots & \dots & \vdots \\ R_{K-1} & R_{K-2} & \dots & R_0 \end{bmatrix}$$

full row rank, i.e., u is persistently exciting of order K .

Consequences for system identification: persistence of excitation

- ▶ Ho-Kalman algorithm $\implies$ system identification algorithm.

Ho-Kalman is the inspiration of subspace methods [De Moor, Larimore, ..]

- ▶ Partial realization theory $\implies$ the first $2n$ Markov parameters are sufficient to recover a realization of order n .

$\{y(t), u(t)\}_{t=1}^T$ is **persistently exciting** $\iff$ we can recover from $\{y(t), u(t)\}_{t=1}^T$ the first $2n$ Markov parameters.

Consequences for identification: identifiability

- ▶ Parameterized family $\{(A(\theta), B(\theta), C(\theta))\}_\theta$ of LTI
- ▶ $Y(\theta)$ input-output map (data) of $(A(\theta), B(\theta), C(\theta))$
- ▶ Y measured data
- ▶ Parametric system identification:

$$\theta_0 = \arg \min_{\theta} \|Y(\theta) - Y\|$$

Problem: if $\theta \mapsto Y(\theta)$ not injective, then multiple optima.

- ▶ $\{(A(\theta), B(\theta), C(\theta))\}_\theta$ called **identifiable**, if $\theta \mapsto Y(\theta)$ injective.

Consequences for identification: identifiability

Theorem 4 (Glover, Willems & J. van den Hof)

Assume

- ▶ $\theta \mapsto (A(\theta), B(\theta), C(\theta))$ is injective.
- ▶ $\forall \theta, (A(\theta), B(\theta), C(\theta))$ minimal (reachable & observable)

Then $\{(A(\theta), B(\theta), C(\theta))\}_\theta$ is identifiable, $\iff$

$\forall \theta_1 \neq \theta_2$, there is no isomorphism between $(A(\theta_1), B(\theta_1), C(\theta_1))$ and $(A(\theta_2), B(\theta_2), C(\theta_2))$

Consequences for identification: identifiability

Theorem 5 (Glover, Willems)

Assume that $(A(\theta_0), B(\theta_0), C(\theta_0))$ is minimal. Then $\{(A(\theta), B(\theta), C(\theta))\}_\theta$ is identifiable for θ in a neighborhood of θ_0 , if

$$M_F(\theta_0) = \begin{bmatrix} \frac{dF}{dT}(I, \theta_0) & \frac{dF}{d\theta}(I, \theta_0) \end{bmatrix},$$

is full column rank, where for any nonsingular matrix T and parameter θ ,

$$F(T, \theta) = (TA(\theta)T^{-1}, TB(\theta), C(\theta)T^{-1})$$

More elaborate ideas: data-driven local coordinates.

Consequences for identification: identifiability

Exercise 10

Consider the scalar SISO family

$$x(t+1) = A(\theta)x(t) + B(\theta)u(t), \quad y(t) = C(\theta)x(t),$$

and

$$F(T, \theta) = (A(\theta), TB(\theta), C(\theta)T^{-1}).$$

with scalar coordinate change $T \neq 0$.

At $\theta_0 = 1$, compute $M_F(\theta_0) = \begin{bmatrix} \frac{dF}{dT}(1, \theta_0) & \frac{dF}{d\theta}(1, \theta_0) \end{bmatrix}$ for

$$A(\theta) = \theta, \quad B(\theta) = 1, \quad C(\theta) = 1$$

and for

$$A(\theta) = 1, \quad B(\theta) = \theta, \quad C(\theta) = \frac{1}{\theta}.$$

Which parametrization satisfies the rank condition in the theorem of Glover, Willems?